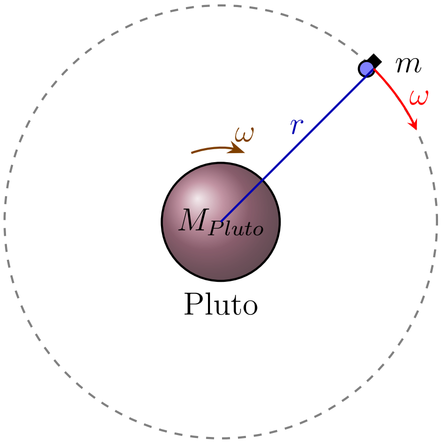
Name: ____________________ Neptun code: ____________________

(KTXFI1EBNF) Physics I. Test 1 Solutions
May 11, 2026

1. At what distance from Pluto does a Pluto-stationary satellite orbit in the plane of the equator if it remains constantly above the same point on Pluto's surface? ($M_{\text{Pluto}} = 1.305 \cdot 10^{22}$ kg, $T_{rot} = 6.38723$ day, $R_{Pluto} = 1188$ km, $G = 6.6743 \cdot 10^{-11}$ Nm^2/kg^2.)



Pluto-stationary orbit in the Equatorial plane

For a stationary orbit, the orbital period of the satellite must be equal to the rotation period of Pluto:

$$T = 6.38723\,\text{days}.$$

Converting this into seconds:

$$T = 6.38723 \cdot 24 \cdot 3600\, s \approx 5.51857 \cdot 10^5\,\text{s}.$$

The gravitational force provides the centripetal force:

$$\frac{GM_{Pluto}m}{r^2} = m\frac{4\pi^2 r}{T^2}.$$

After simplifying and solving for the orbital radius:

$$r = \sqrt[3]{\frac{GM_{Pluto}T^2}{4\pi^2}}.$$

Substituting the data:

$$r = \sqrt[3]{\frac{(6.6743 \cdot 10^{-11}\, Nm^2/kg^2)(1.305 \cdot 10^{22}\, kg)(5.51857 \cdot 10^5\, s)^2}{4\pi^2}}.$$

$$r \approx 1.88 \cdot 10^7\,\text{m}.$$

Therefore, the

$$r \approx 1.88 \cdot 10^4\,\text{km}$$

from the center of Pluto.
Since the radius of Pluto is approximately

$$R_{\text{Pluto}} \approx 1188\,\text{km},$$

the altitude above the surface is

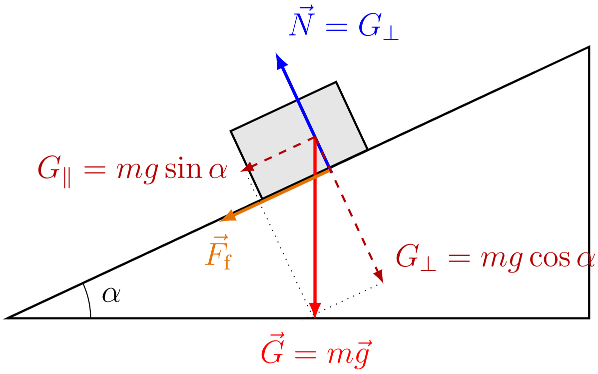
$$h = r - R_{\text{Pluto}} \approx 1.88 \cdot 10^4 - 1.19 \cdot 10^3.$$

$$\boxed{h \approx 1.76 \cdot 10^4\,\text{km}}$$

(10 points)

2. A 5 kg object slides down a 4-meter-high incline that makes an angle of 45° with the horizontal. What is the speed of the object when it reaches the bottom of the incline if it starts from rest at the top and:
   (a) there is no friction between the object and the incline?
   (b) the coefficient of friction is 0.1?

   ($g = 9.81\,\mathrm{m/s^2}$)



The height of the incline is

$$h = 4\,\mathrm{m}.$$

(a) Without friction, mechanical energy is conserved:

$$mgh = \frac{1}{2}mv^2.$$

Therefore,

$$v = \sqrt{2gh}.$$

Substitution gives

$$v = \sqrt{2 \cdot 9.81\,m/s^2 \cdot 4\,m} \approx \boxed{8.9\,\mathrm{m/s}.}$$

(b) With friction, the friction force is

$$F_f = \mu mg\cos\alpha.$$

The length of the incline is

$$s = \frac{h}{\sin\alpha}.$$

The work done by friction is

$$W_f = F_f s = \mu mg\cos\alpha\frac{h}{\sin\alpha} = \mu mgh\cot\alpha.$$

Using energy conservation with energy loss:

$$mgh - \mu mgh\cot\alpha = \frac{1}{2}mv^2.$$

Thus,

$$v = \sqrt{2gh(1-\mu\cot\alpha)}.$$

For

$$\alpha = 45°, \qquad \cot 45° = 1,$$

we get

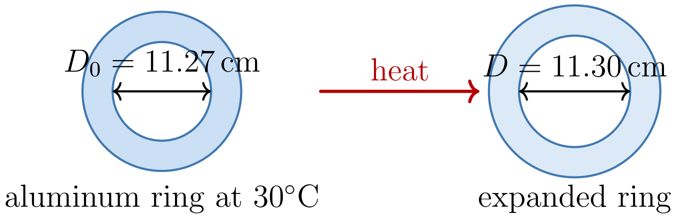
$$v = \sqrt{2gh(1-\mu)}.$$

Substitution:

$$v = \sqrt{2 \cdot 9.81\,\mathrm{m/s^2} \cdot 4\,\mathrm{m} \cdot (1-0.1)} \approx \boxed{8.4\,\mathrm{m/s}}.$$

(10 points)

3. At a temperature of 30°C, a steel shaft with a diameter of 11.30 cm needs to be fitted with an aluminum ring with an inner diameter of 11.27 cm at the same temperature. The linear thermal expansion coefficient of steel is $12 \cdot 10^{-6}\,\mathrm{K}^{-1}$, and the linear thermal expansion coefficient of aluminum is $28.7 \cdot 10^{-6}\,\mathrm{K}^{-1}$. What temperature must the ring be heated to?



The aluminum ring must expand from

$$D_0 = 11.27\,\mathrm{cm} = 0.1127\,\mathrm{m}$$

to

$$D = 11.30\,\mathrm{cm} = 0.113\,\mathrm{m}.$$

The linear thermal expansion formula is

$$D = D_0(1 + \alpha \Delta T).$$

Thus,

$$\Delta T = \frac{D - D_0}{\alpha D_0}.$$

Substitution gives

$$\Delta T = \frac{0.113\,\mathrm{m} - 0.1127\,\mathrm{m}}{(2.87 \cdot 10^{-5}\,1/\mathrm{K}) \cdot 0.1127\,\mathrm{m}} \approx 92.75\,\mathrm{K}.$$

The initial temperature is

$$T_0 = 30°\mathrm{C}.$$

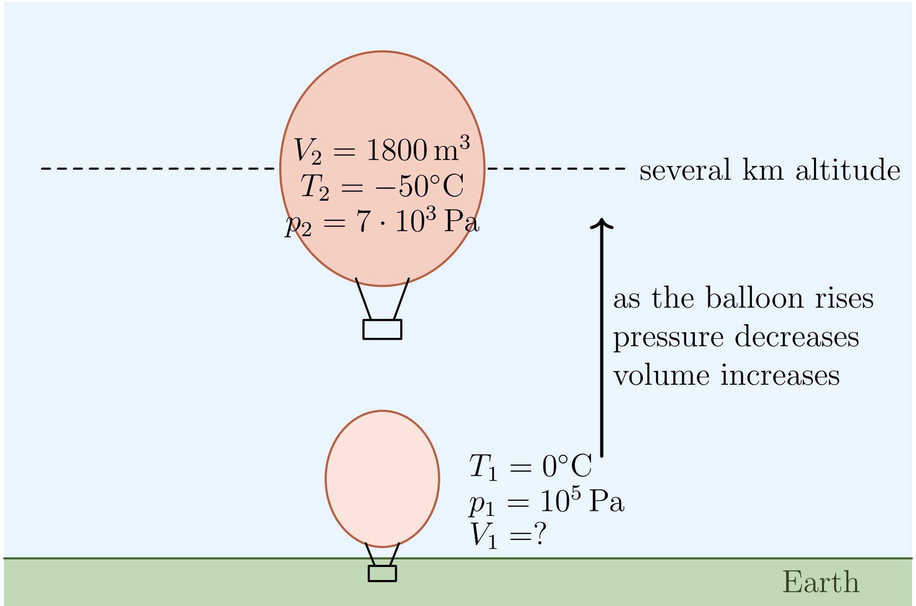
Therefore,

$$T = T_0 + \Delta T = 30°\mathrm{C} + 92.75°\mathrm{C} \approx \boxed{122.75°\mathrm{C}.}$$

(10 points)

4. A weather balloon floating at an altitude of several kilometers has a volume of $1800\,\mathrm{m}^3$. The temperature and pressure of the helium inside it are the same as those of the surroundings: $-50°\mathrm{C}$ and $7\cdot 10^3\,\mathrm{Pa}$, respectively.
   (a) How many moles of helium are in the balloon?
   (b) What is the mass of the helium gas? $\left(M = 4\,\frac{\mathrm{g}}{\mathrm{mol}}\right)$
   (c) What was the volume of the balloon when it was launched from Earth, if the gas inside the balloon was characterized by a temperature of $0°\mathrm{C}$ and a pressure of $10^5\,\mathrm{Pa}$?

   $(R = 8.31\,\mathrm{J/K})$



Given:

$$V = 1800\,\mathrm{m}^3, \qquad p = 7\cdot 10^3\,\mathrm{Pa}, \qquad T = -50°\mathrm{C} = 223.15\,\mathrm{K}.$$

(a) From the ideal gas law,

$$pV = nRT.$$

Thus,

$$n = \frac{pV}{RT}.$$

Substitution:

$$n = \frac{(7\cdot 10^3\,\mathrm{Pa})(1800\,\mathrm{m}^3)}{8.314\,\mathrm{J/K}\cdot 223.15\,\mathrm{K}} \approx \boxed{6.8\cdot 10^3\,\mathrm{mol}}.$$

(b) The molar mass of helium is

$$M = 4\,\mathrm{g/mol} = 0.004\,\mathrm{kg/mol}.$$

The mass is

$$m = nM = (6.79\cdot 10^3\,\mathrm{mol})(0.004\,\mathrm{kg/mol}) \approx \boxed{27\,\mathrm{kg}}.$$

(c) At launch:

$$T_0 = 0°\mathrm{C} = 273.15\,\mathrm{K}, \qquad p_0 = 10^5\,\mathrm{Pa}.$$

Using the same amount of gas:

$$\frac{pV}{T} = \frac{p_0 V_0}{T_0}.$$

Thus,

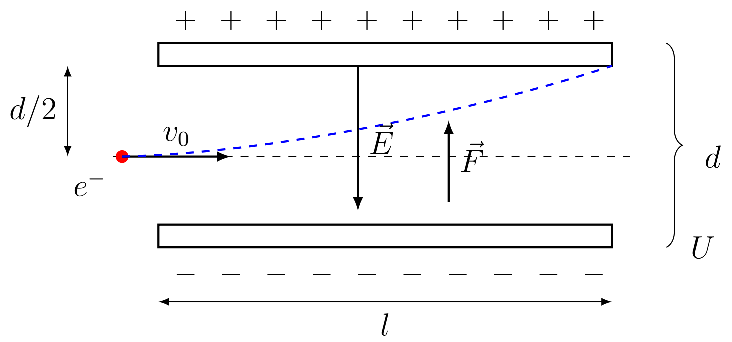
$$V_0 = \frac{pVT_0}{p_0 T}.$$

Substitution:

$$V_0 = \frac{pVT_0}{p_0 T} = \frac{(7\cdot 10^3\,\mathrm{Pa})(1800\,\mathrm{m}^3)(273.15\,\mathrm{K})}{(10^5\,\mathrm{Pa})(223.15\,\mathrm{K})} \approx \boxed{1.54\cdot 10^2\,\mathrm{m}^3}.$$

(10 points)

5. An electron enters the region between the parallel plates shown in the figure halfway between the plates. The length of the plates is $l = 12\,\text{cm}$, their separation is $d = 2\,\text{cm}$, and the voltage between them is $U = 300\,\text{V}$. What must the initial velocity of the electron be so that it just exits the region between the plates? ($m_e = 9.1 \cdot 10^{-31}\,\text{kg}$, $e = 1.6 \cdot 10^{-19}\,C$)



Given:

$$l = 12\,\text{cm} = 0.12\,\text{m}, \qquad d = 2\,\text{cm} = 0.02\,\text{m}, \qquad U = 300\,\text{V}.$$

The electric field between the plates is

$$E = \frac{U}{d}.$$

Thus,

$$E = \frac{300}{0.02} = 15000\,\text{V/m}.$$

The electric force acting on the electron is

$$F = eE.$$

Therefore, the acceleration of the electron is

$$a = \frac{F}{m_e} = \frac{eE}{m_e}.$$

Substituting the values:

$$a = \frac{(1.602 \cdot 10^{-19}\,\text{C})(15000\,\text{V/m})}{9.109 \cdot 10^{-31}\,\text{kg}} \approx 2.637 \cdot 10^{15}\,\text{m/s}^2.$$

The electron enters halfway between the plates, therefore it must move vertically by

$$y = \frac{d}{2} = 0.01\,\text{m}$$

in order to just reach the upper plate at the exit.
The vertical motion is uniformly accelerated:

$$y = \frac{1}{2}at^2.$$

Thus,

$$0.01\,\text{m} = \frac{1}{2}(2.637 \cdot 10^{15}\text{m/s}^2)t^2.$$

Solving for the time:

$$t = \sqrt{\frac{2 \cdot 0.01\,\text{m}}{2.637 \cdot 10^{15}\,\text{m/s}^2}} \approx 2.754 \cdot 10^{-9}\,\text{s}.$$

The horizontal motion is uniform:

$$l = v_0 t.$$

Therefore,

$$v_0 = \frac{l}{t}.$$

Substituting the values:

$$v_0 = \frac{l}{t} = \frac{0.2\,\mathrm{m}}{2.754 \cdot 10^{-9}}\,\mathrm{s} \approx 4.358 \cdot 10^7\,\mathrm{m/s}.$$

Therefore, the required initial velocity is

$$\boxed{v_0 \approx 4.36 \cdot 10^7\,\mathrm{m/s}}$$

(10 points)

Requirement for obtaining the course signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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